

Optimising the Performance of LLM Chatbots: Multi-Agent Evolutionary Prompt Optimisation for Chatbots (MAEPOC)

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Abstract

As large language model (LLM)-based chatbots gain traction in role-specific domains, there is a growing demand for automated prompt optimisation frameworks that deliver consistent, high-quality, and interpretable interactions. We present **MAEPOC** (Multi-Agent Evolutionary Prompt Optimisation for Chatbots), a novel framework that combines evolutionary algorithms with coordinated LLM agents to iteratively refine chatbot prompts. A key innovation of MAEPOC is its use of structured, domain-specific rubrics—repurposed from expert evaluation standards—to guide prompt evolution through interpretable, task-aligned scoring. The framework also employs an LLM-based user simulator, enabling fully automated optimisation without the need for human-labelled data.

We evaluate MAEPOC in a medical pre-consultation setting using a rubric adapted from the Master Interview Rating Scale (MIRS). MAEPOC consistently outperforms both PromptGPT and the initial prompt seed by 33% and 13%, respectively, in average rubric scores, and achieves the best overall performance across multiple independent runs. Ablation studies confirm that both reflection (memory of past prompt–fitness pairs) and increased sampling entropy contribute to improved results, validating key design choices in the framework. An interpretability analysis further reveals that No-Rubric LLM-Judges, despite occasionally producing high raw scores, exhibits reduced domain alignment. Together, these results demonstrate that MAEPOC can generate high-performing prompts while maintaining transparency and reliability—underscoring the value of rubric-guided, multi-agent optimisation for safe and effective domain-specific chatbot development.

1 Introduction

Background

Large Language Model (LLM)-based chatbots have encountered a huge rise in popularity in recent years, highlighting the need for a framework to reliably optimize prompts tailored for each chatbot’s roles automatically. LLM-based chatbots have revolutionized human-computer interaction by leveraging vast corpora of text data and advanced transformer architectures. The emergence of models such as OpenAI’s ChatGPT and Google’s Bard has demonstrated the potential of LLMs to generate coherent, context-aware, and human-like responses in real time [2, 16]. These chatbots are increasingly integrated into customer service, education, and creative industries, underscoring a shift from rule-based systems to data-driven, generative AI agents [17].

Challenges

A chatbot’s environment differs significantly from traditional LLM tasks such as question-answering and natural language inference. The evaluation of a chatbot’s performance is also human-subjective

and encompasses many different factors, such as tone of response. Such implicit nuances in the human language are difficult to encode.

It follows that evaluating LLMs tasked with role-specific interactions, such as medical chatbots, is inherently complex due to the lack of standardized, transferable performance metrics. The evaluation of chatbot prompts is role-dependent and there exists no universal metric. For instance, a banking chatbot trivially serves a different objective than medical chatbots and requires different performance metrics.

A fair amount of research has been conducted for automatic prompt optimisation techniques [19] but to our knowledge, no frameworks have addressed any methodology for evaluating LLM-based chatbots. Proposed strategies for traditional LLM tasks include model fine-tuning, genetic algorithms and generative adversarial methods [19]. Crucially, these strategies have largely only been demonstrated in environments where the outcomes can be evaluated quantitatively.

Existing prompt optimisation techniques, although effective in constrained settings, struggle in interactive, open-ended tasks where success depends on contextual, role-specific performance rather than lexical similarity or general-purpose preferences. Evolutionary approaches have recently gained attention for prompt discovery, but most rely on token-level edit heuristics, lack interpretability, or require pre-defined reward functions. Similarly, LLM-as-a-Judge methods suffer from inconsistency and lack of grounding, making them unreliable for use in automated optimisation loops.

To this end, we propose capitalising on the natural language processing and reasoning capabilities of LLMs to perform the evaluation of exchanges between user and chatbot in a quantitative manner based on a rubric. We also utilise a separate LLM to mimic the user’s interactions with the chatbot to remove human subjectivity and enable automation. We propose the use of evolutionary algorithms (EAs) within our framework, specifically LLM-powered EAs to drive our optimisation algorithms since LLMs have recently proven to be effective optimisers in EAs [12].

Our Contributions

In this work, we present MAEPOC (Multi-Agent Evolutionary Prompt Optimisation for Chatbots), a novel framework that addresses these limitations. Our key contributions are as follows:

1. A multi-agent, automatic evolutionary framework for prompt optimisation that operates in black-box settings and requires no ground-truth labels, leveraging LLMs for mutation, crossover, evaluation, and user simulation.
2. A domain-specific evaluation methodology that repurposes structured human rubrics (e.g. MIRS) for LLM-based scoring, enabling consistent and interpretable fitness evaluation aligned with real-world task objectives.
3. A novel use of LLM-as-evaluator grounded in rubric criteria, addressing prior issues of scoring ambiguity in LLM feedback-based assessment.
4. Population-level optimisation: Unlike many existing approaches that focus on identifying a single optimal prompt (e.g., PromptGPT or Self-Instruct), MAEPOC evolves and maintains an entire population of high-performing prompts. This allows for diversity and multiple deployment options depending on task constraints or user preferences.

5. Demonstration of effectiveness in a medical pre-consultation chatbot setting, where our framework successfully evolves prompts that achieve better evaluation scores.

2 Related Works

In this section, we will examine literature related to our work. First, we will look at existing methods to evaluate prompt candidates. Secondly, we will look at existing prompt optimisation techniques and how they can be incorporated into our framework. Then, we will dive into the use of optimisation operators in EAs. Finally, we will list our contributions towards designing an automatic prompt optimisation framework, built on existing literature, to fill the gaps identified in automatic prompt design for chatbots.

2.1 Evaluation Methods

Crucial to our framework is an evaluation method to identify strong prompt candidates that we can use as a basis for improvement. In the context of evaluating chatbot exchanges, there is extensive literature covering methodologies that could potentially be incorporated. The methodologies can be classified into numeric score feedback, LLM feedback and human feedback [19]. We will focus on the first two methodologies keeping to our goal of automatic prompt optimisation.

Numeric Score Feedback

Traditional evaluation metrics for LLMs, such as ROUGE [15] and BLEU [18], rely on lexical overlap between generated outputs and reference texts. While effective for tasks like summarization or translation, these metrics primarily capture surface-level similarity and fail to assess the functional quality of prompts in interactive settings. In contrast, MAEPOC introduces a functional evaluation paradigm: prompts are scored based on their ability to elicit desirable behaviors—such as extracting clinically relevant symptom information—in role-specific dialogues like medical pre-consultations.

To address the limitations of overlap-based metrics, recent research has proposed reward-model scoring systems that use learned models to assign scores to prompt-response pairs based on human preference. One such model, RewardAnything, generalizes across diverse instruction-following tasks by capturing latent reward signals aligned with human feedback [26]. However, while these models improve upon purely lexical metrics, they are trained on broad, general-purpose preference data and lack grounding in task-specific, domain-aware objectives. MAEPOC addresses this gap by incorporating structured, rubric-based evaluation aligned with professional standards, enabling fine-grained assessment of prompt effectiveness in specialized domains.

LLM Feedback

The LLM-as-a-Judge paradigm is not new, having been adopted in several recent works to evaluate LLM outputs across a range of tasks [27]. This includes the use of LLM-generated feedback to score chatbot responses, as seen in MT-Bench and related evaluation frameworks. However, a key limitation of existing LLM feedback methodologies is the ambiguity and opacity of their scoring rationale. Although techniques such as chain-of-thought prompting have been employed to encourage LLM judges to justify their decisions [24, 25], the underlying basis for assigning a particular score remains unclear and often inconsistent.

This ambiguity poses a significant challenge for automatic prompt optimisation frameworks like MAEPOC, where consistency and comparability of evaluation across generations is critical. LLM judges may apply shifting standards or subjective biases across different prompts or conversations, undermining the reliability of the feedback signal. MAEPOC addresses this shortcoming by grounding the LLM-Assessor’s evaluations in explicit, domain-specific human rubrics—such as the Master Interview Rating Scale (MIRS)—thus ensuring a consistent, interpretable, and task-aligned scoring methodology.

2.2 Prompt Optimisation (PO) Techniques

Prompt optimisation has emerged as a central research focus in enhancing large language model (LLM) performance across diverse tasks. Early work primarily relied on manual prompt engineering, where practitioners iteratively refined prompts through heuristics and task-specific insight. Although effective in some cases, this approach is time-consuming, brittle, and often lacks generalizability.

To overcome these limitations, a number of automated prompt optimisation techniques have been proposed. Among gradient-based methods, Prompt Tuning [10] and Prefix Tuning [11] learn continuous soft prompts by updating trainable embeddings prepended to frozen model parameters. These approaches show strong performance but require white-box access to model internals and gradients, making them infeasible for most commercial API-based models.

For black-box settings, AutoPrompt [21] introduced a discrete prompt search method that replaces individual tokens in a prompt to maximize task performance using only model outputs. Similarly, PromptSource [1] provides a templated interface for prompt design, allowing for community-driven prompt experimentation across benchmark datasets. However, these methods typically assume the existence of reference outputs or labeled data, which are not available in open-ended dialogue tasks like medical pre-consultations.

Reinforcement learning-based methods such as RLPrompt [4] further explore the prompt space by treating the LLM as an environment and learning optimal prompts via reward signals. Adversarial prompt learning approaches such as PromptAttack [8] generate prompts that stress-test or exploit LLM vulnerabilities. While these methods offer stronger optimisation capabilities, they often rely on well-defined scalar reward functions and are difficult to adapt to subjective, multi-turn interaction tasks.

In contrast, MAEPOC targets role-based, functional prompt effectiveness by directly measuring how well a prompt elicits target behaviors during simulated interactions. Rather than relying on static labels or reference responses, prompts are evaluated based on the quality of the resulting user-chatbot dialogue, scored using structured domain-specific rubrics. This positions MAEPOC as a suitable solution for black-box, interaction-heavy tasks where conventional prompt optimisation metrics and techniques fall short.

2.3 Automatic PO Techniques

Recent research has increasingly focused on fully automated prompt optimisation methods that treat prompt design as a search or learning problem. These approaches aim to discover high-performing prompts without human intervention, making them particularly attractive for scalable

deployment in real-world systems.

One major class of methods formulates prompt optimisation as a search problem in discrete space. PromptBench [5] introduced a suite of benchmarks and prompt search algorithms (e.g., greedy, beam, and random search) for optimizing prompts across classification and generation tasks, evaluating them based on task-specific performance. PromptGPT [7] uses large language models in a self-play loop to generate, evaluate, and refine prompts through bootstrapped natural language feedback. While promising, both approaches often rely on predefined evaluation datasets or reference answers, which can limit their applicability to open-ended dialogue tasks.

Another class of methods leverages LLMs as meta-optimisers. Works such as Self-Instruct [23] and InstructZero [6] use LLMs to generate both prompts and task instances iteratively, thereby constructing their own training data and instructions. While powerful, these methods still typically assume a clear task formulation or labeled outputs to assess prompt effectiveness.

More recently, multi-agent and evolutionary frameworks have been explored for automatic prompt discovery. PE2 [14] and LM-Debugger [28] involve agents playing generator, evaluator, or explainer roles, but often rely on manually curated prompt pools or specific task ground truths. However, the success of these methods still hinges on the availability of clear, generalizable reward signals, which are difficult to define in domains like medical dialogue.

MAEPOC builds on this line of work by introducing a multi-agent, rubric-driven evolutionary framework specifically tailored to open-ended, role-specific chatbot tasks. Unlike prior methods that rely on task-specific references or learned reward models, MAEPOC uses a structured domain rubric (e.g., the Master Interview Rating Scale) and a dedicated LLM evaluator to guide optimisation, enabling scalable and interpretable prompt refinement in complex dialogue settings without manual supervision.

2.4 EA Optimisation Operators

EAs have been widely used for optimisation in discrete, continuous, and structured domains due to their flexibility and gradient-free nature. In the context of prompt optimisation, EAs offer a compelling framework for exploring large, non-differentiable prompt spaces without requiring access to model internals.

Conventional EAs consist of three primary operators: selection, mutation, and crossover. Traditionally, candidate solutions are encoded as strings (e.g., binary or token sequences), and the fitness of each candidate is evaluated based on a task-specific objective function. Mutation introduces local variability by randomly altering elements of a candidate, while crossover combines information from two parent solutions to generate new candidates. Selection mechanisms (e.g., tournament, rank, or fitness-proportional) determine which candidates survive or reproduce in the next generation.

Recent work has adapted EAs for natural language settings. For example, PromptEvo [3] applies token-level mutation and crossover to prompt tokens in instruction tuning, while EvoPrompt [29] employs edit-based operations on prompt templates to optimise question-answering prompts. However, these approaches typically require carefully designed token-level heuristics and hand-tuned edit operators. Moreover, the search space remains brittle due to linguistic constraints, and evaluation still depends on static labels or reference outputs.

To overcome these limitations, recent research has begun using LLMs as evolutionary operators. For instance, Liu et al. [13] propose using LLMs to perform natural language-based crossover and mutation, thereby avoiding token-level corruption and maintaining semantic fluency. This approach allows for black-box, language-native evolution, where LLMs themselves propose and modify new candidate prompts based on prior generations.

MAEPOC builds on this line of work by introducing a multi-agent evolutionary framework where all evolutionary operations—mutation, crossover, and evaluation—are LLM-based.

3 Problem Formalisation

We now formalise MAEPOC as a black-box, multi-agent evolutionary optimisation algorithm over natural language prompts.

Prompt Space and Objective

Let $\mathcal{P}$ denote the space of valid prompts over a vocabulary $\mathcal{V}$. Each prompt $p \in \mathcal{P}$ is a token sequence:

$$p = (w_1, w_2, \dots, w_n), \quad w_i \in \mathcal{V}, \quad n \leq N_{\max}$$

Let $f : \mathcal{P} \rightarrow \mathbb{R}$ denote a black-box fitness function representing the quality of prompt p in eliciting desirable chatbot behavior during interaction.

Since $f(p)$ is expensive and stochastic, we approximate it using Monte Carlo sampling:

$$\hat{f}(p) = \frac{1}{k} \sum_{i=1}^k f_i(p), \quad f_i(p) \sim \text{RubricEval}(\text{SimulateChat}(p))$$

where $\text{SimulateChat}(p)$ produces a conversation from prompt p using LLM-US, and $\text{RubricEval}(\cdot)$ scores the conversation using an LLM-A grounded in a domain rubric.

Our goal is to find an optimal prompt:

$$p^* = \arg \max_{p \in \mathcal{P}} \hat{f}(p)$$

Agents and Operator Functions

Let $\mathcal{A} = \{A_{\text{US}}, A_{\text{A}}, A_{\text{EO}}\}$ denote the three LLM-based agents used in the framework:

- A_{US} : User Simulator — generates user responses given a prompt and chat history.
- A_{A} : Assessor — evaluates completed user-chatbot conversations using a rubric $\mathcal{R}$ and outputs a scalar fitness score.
- A_{EO} : Evolutionary Optimizer — applies mutation and crossover operators to produce new prompt candidates.

We define two stochastic language-native operators parameterized by A_{EO} :

$$\begin{aligned} \text{Mutation: } \mu : \mathcal{P} &\rightarrow \mathcal{P}, \quad \mu(p) \sim A_{\text{EO}}(p \mid \text{ReflectionContext}) \\ \text{Crossover: } \chi : \mathcal{P} \times \mathcal{P} &\rightarrow \mathcal{P}, \quad \chi(p_1, p_2) \sim A_{\text{EO}}(p_1, p_2 \mid \text{ReflectionContext}) \end{aligned}$$

Evolutionary Process

Let $\mathcal{D}_t = \{p_1^t, p_2^t, \dots, p_N^t\}$ denote the population of N prompts at generation t . Each evolutionary epoch proceeds as follows:

1. **Selection:** Select prompts from $\mathcal{D}_t$ using heat-weighted sampling based on $\hat{f}(p)$.

2. **Variation:**

$$p_{\text{child}} \sim \begin{cases} \mu(p) & \text{w.p. } 1 - \rho \\ \chi(p, p') & \text{w.p. } \rho \end{cases}$$

where $\rho \in [0, 1]$ is the crossover rate.

3. **Evaluation:** Estimate fitness $\hat{f}(p_{\text{child}})$ using $A_{\text{A}}(\text{SimulateChat}(p_{\text{child}}))$.

4. **Replacement:** If $\hat{f}(p_{\text{child}})$ exceeds the lowest fitness in $\mathcal{D}_t$, replace that prompt.

5. **Reseeding:** If no improvement is observed for T generations and convergence criteria are met, reseed:

$$\mathcal{D}_{t+1} \leftarrow \text{ReseedWithVariants}(\arg \max_{p \in \mathcal{D}_t} \hat{f}(p))$$

This iterative process continues until a fixed number of generations is reached.

Summary

MAEPOC formalises prompt optimisation as a stochastic, population-based search over $\mathcal{P}$ guided by multi-agent interactions:

$$\hat{f}(p) \approx A_{\text{A}}(\text{SimulateChat}(p, A_{\text{US}}))$$

This setup allows for robust, interpretable, and domain-specific prompt discovery using only a rubric and seed prompt, without requiring access to ground-truth data or model internals.

4 Proposed Framework

MAEPOC consists of 4 LLM-based agents (refer to figure 1). Firstly, we require the LLM chatbot whose prompt we would like to optimise. A population of prompts are stored in a prompt container (PC), where candidate prompts will be stored. Secondly, we introduce the LLM User Simulator (LLM-US), which simulates a user interacting with the chatbot. Thirdly, the LLM Assessor (LLM-A) which is tasked with evaluating the resulting exchange and quantifying its quality. Lastly, an LLM Evolutionary Optimizer (LLM-EO) to execute the mutation/crossover of select candidate prompts from within the PC. Note that all LLMs described in this paper utilise OpenAI’s ChatGPT-4o models.

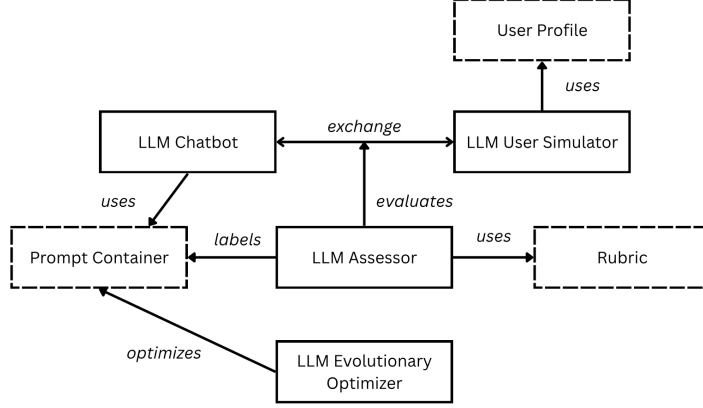


Figure 1: MAEPOC Framework

4.1 LLM-Assessment

Our framework incorporates a Large Language Model Assessor (LLM-A) to evaluate user-chatbot interactions. The LLM-A uses a predefined rubric to perform structured, criteria-based evaluation. This novel application leverages modern LLMs’ natural language understanding and reasoning capabilities to generate human-aligned assessments. Providing the LLM-A with a well-defined rubric allows it to quantify an exchange’s performance in a transparent way.

We define the structure of a rubric as:

- $\mathcal{A} = \{A_1, A_2, \dots, A_X\}$ — the set of evaluation areas.
- $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ — the set of possible scores (e.g., $\{0, 1, 2, 3\}$).
- $\mathcal{R}_{A_i} = \{r_1^{(i)}, r_2^{(i)}, \dots, r_N^{(i)}\}$ — the set of criteria corresponding to scores for area A_i .

$$\text{Rubric} = \{(A_i, s_j, \mathcal{R}_{A_i}) \mid A_i \in \mathcal{A}, s_j \in \mathcal{S}\}$$

A valid rubric that satisfies the defined structure can be inserted into the LLM-A prompt template. The prompt template provided to the LLM-A is as follows:

```

You are a strict, critical assessor tasked with evaluating an exchange
between a chatbot and user.
You are to base your evaluation strictly on the marking rubric below.
For each area, walk through the reasoning behind assigning a score based on
the given exchange.
Rubric: <RUBRIC>
Exchange: <EXCHANGE>
Break down the scores, justifying why each area is assigned its score.
Based on the breakdown, sum up the individual scores for each area.
Strictly output only a number.

```

An example rubric follows:

```
< Area | 0 | 1 | 2 | 3 >
< Clarity | Writing is unclear and disorganized |
Some organization but lacks clarity | Mostly clear and well-structured |
Exceptionally clear and well-organized > < Creativity | Lacks original
thought | Somewhat creative | Generally creative with original ideas |
Highly creative and original throughout >
```

4.2 LLM-User Simulation

To reduce human subjectivity from the creation of user replies, we introduce the LLM-US to simulate a user interacting with the chatbot. The LLM-US is provided with a user-profile and user-chatbot exchange history to generate a simulated response. The prompt provided to our LLM-US is as follows:

```
You are an assessor tasked with simulating a user's response, in an
exchange with a chatbot.
You are to base the information you provide strictly on the simulated
profile below, as well as the exchange history with the chatbot.
You should provide only 1 response at a time.
Strictly output only a response (no sender label needed).
Profile: <PROFILE>
Exchange: <EXCHANGE>
```

4.3 LLM-Evolutionary Optimisation

In our framework, the generation of new prompts is done via LLM-EOs. LLM-EOs avoid the need for token-level scrutinisation when performing mutations and crossovers, significantly increasing the ease of implementation. A fitness score is assigned to the children candidates and if sufficiently high, is inserted into the population of existing solutions, replacing existing solutions. This process can be repeated until a population of high-performing candidates is obtained. The use of LLMs as evolutionary optimisers have been studied extensively [12], hence we will not dwell too deep into the topic in this paper, instead describing the surface-level implementation of EAs in the MAEPOC framework.

Population Seeding

MAEPOC requires a prompt to serve as a seed for future generations. With the seed, we instruct an LLM to generate multiple variations of the prompt, which will serve as the initial population of candidate solutions. The initial population is stored in the PC. The prompt is provided as follows:

```
Generate a new prompt based on the prompt provided.
The new prompt should perform a similar function as the original prompt.
The new prompt should be more descriptive than the original prompt.
Identify the strengths of the original prompt.
Build on the strengths of the original prompt.
Reflect on how improvements can be made to the original prompt.
The new prompt should implement these improvements on the original prompt.
The new prompt should begin with "You are a...".
Strictly output only the new prompt's content.
Given prompt: <SEED>
```

Candidate Selection

We now select prompts for mutation and crossovers. In EAs, this process is built on the ‘survival of the fittest’ presumption, which suggests that fitter parent candidates are more likely to result in better offspring candidates. For mutations, this entails mutating the fittest prompt. For crossovers, this entails combining the two fittest prompts. However, this occasionally leads to local optima resulting in the algorithm repeatedly failing to generate a fitter prompt, presumably because there exists no fitter prompt that can be created by evolving the existing. To prevent this, we include a HEAT hyperparameter (refer to table ??) which encourages random selection of candidates for mutation and crossovers. For crossovers, the fittest candidate is selected but the mating candidate is randomly selected from the population.

Reflection

To enhance the quality of generated prompts, we incorporate a reflection mechanism into the mutation and crossover process. Specifically, we provide the LLM-EO with context in the form of previously evaluated prompt–fitness score pairs. This allows the model to reflect on what characteristics are associated with higher or lower performance and generate new prompts that build on prior strengths. The input to the LLM-EO includes up to N prompts from the current population, formatted as follows:

```
Reflect on why the prompts below are associated with their scores.  
Prompt: <PROMPT_1>, Fitness: <FITNESS_1>  
Prompt: <PROMPT_2>, Fitness: <FITNESS_2>  
...  
Prompt: <PROMPT_N>, Fitness: <FITNESS_N>
```

Following this reflection prompt, the LLM-EO is instructed to either mutate a single candidate or perform a crossover between two candidates, using the context to guide its transformation decisions. This approach encourages the model to internalise prompt–performance associations and apply them during evolution, improving the likelihood of producing high-fitness offspring.

Crossovers

Crossovers involve combining two candidate prompts by passing them through a crossover operator. We provide the LLM-EO with two selected candidates as input. The LLM-EO then produces an offspring prompt. The crossover prompt provided to the LLM-EO, which includes the reflection prompt described in section 4.3, is as follows:

```
<REFLECTION>  
Applying what you have reflected upon, combine the following prompts  
into one.  
The new prompt should begin with 'You are a'.  
Strictly output only the new prompt.  
1: <PROMPT_1>  
2: <PROMPT_2>
```

Mutation

Mutation involves passing a single candidate prompt through a mutation operator, modifying it to obtain an offspring candidate. We provide the LLM-EO with the selected candidate as input, which then produces an offspring prompt. The mutation prompt provided to the LLM-EO, which includes the reflection prompt described in section 4.3, is as follows:

```
<REFLECTION>
Applying what you have reflected upon, modify the following prompt.
Improve the quality of the prompt.
Add relevant specifics while not changing role requirements.
The new prompt should begin with 'You are a'.
Strictly output only the new prompt.
Prompt: <PROMPT>
```

4.4 Reseeding

To escape local optima during optimisation runs, we utilise a reseeding mechanism that periodically repopulates the prompt population with variations of the current fittest candidate. Reseeding is triggered based on two criteria: a convergence window, which monitors whether the average fitness of the population has plateaued, and a patience threshold, which measures how many generations have elapsed without any improvement in the maximum fitness. If both conditions are met, and the theoretical maximum fitness has not yet been reached, the population is reseeded. This strategy enables the search to recover from stagnation and promotes broader exploration of the prompt space.

4.5 Stochastic Variance

During a MAEPOC optimisation run, each child prompt is evaluated only once per generation to estimate fitness. This strategy is consistent with modern stochastic optimisation methods in black-box and evolutionary settings, where computational efficiency and search breadth are prioritised over per-step evaluation accuracy [9, 20]. Repeated evaluations at each step would impose substantial computational overhead without significantly improving optimisation direction. Instead, MAEPOC relies on population-level selection and multi-generational evolution to naturally attenuate noise and promote robust candidates. Final prompts are re-evaluated multiple times post-optimisation to obtain reliable estimates of quality and variance (see tables 2 and 3).

4.6 Pseudocode

Algorithm 1 MAEPOC Prompt Optimisation Framework

```
1:  $P \leftarrow \text{InitialisePopulation}(\text{SeedPrompt})$ 
2:  $\text{best\_fitness} \leftarrow \max(\text{Evaluate}(P))$ 
3:  $\text{epochs\_since\_improvement} \leftarrow 0$ 
4: for  $\text{epoch} = 1$  to  $EPOCHS$  do
5:   Draw random number  $r \in [0, 1]$ 
6:   if  $r < CROSSOVER$  then ▷ Perform crossover
7:      $p_1 \leftarrow$  fittest candidate in  $P$ 
8:      $p_2 \leftarrow$  random candidate in  $P$  (with heat-weighted sampling)
9:      $\text{child} \leftarrow \text{Crossover}(p_1, p_2)$ 
10:  else ▷ Perform mutation
11:     $p \leftarrow$  candidate from  $P$  (fittest with probability  $1 - HEAT$ , else random)
12:     $\text{child} \leftarrow \text{Mutate}(p)$ 
13:  end if
14:   $\text{child\_fitness} \leftarrow \text{Evaluate}(\text{child})$ 
15:  if  $\text{child\_fitness} >$  lowest fitness in  $P$  then
16:    Replace worst candidate in  $P$  with  $\text{child}$ 
17:  end if
18:  if  $\text{child\_fitness} > \text{best\_fitness}$  then
19:     $\text{best\_fitness} \leftarrow \text{child\_fitness}$ 
20:     $\text{epochs\_since\_improvement} \leftarrow 0$ 
21:  else
22:     $\text{epochs\_since\_improvement} \leftarrow \text{epochs\_since\_improvement} + 1$ 
23:  end if
24:  if  $\text{epochs\_since\_improvement} > PATIENCE$  and  $\text{Converged}(P, WINDOW)$  then
25:     $P \leftarrow \text{ReseedWithVariants}(\text{Fittest}(P))$ 
26:     $\text{epochs\_since\_improvement} \leftarrow 0$ 
27:  end if
28: end for
29: return fittest candidate in  $P$ 
```

5 Experiment Set-up

5.1 Context

In any clinical visit, doctors must first ask patients a series of questions to put together a patient demographic before making an informed medical diagnosis. Medical Pre Consultation chatbots can significantly reduce clinic waiting times [22], as doctors no longer have to collect symptom details from patients during medical consultations. This results in an increase in productivity and reduces strain on clinical resources.

Prompt Seed

We constructed a prompt seed by outlining the requirements of a medical pre-consultation chatbot. When evaluated by the LLM-A and LLM-US, it averaged a score of about 25 (refer to table 4). The prompt is provided in annex D.

LLM-A Rubric Metrics

In the context of a pre-consultation chatbot, we base our evaluation metrics in this paper on the Master Interview Rating Scale (MIRS) to mirror educational best practices. MIRS is a common metric used in the medical industry to consistently evaluate patient-interviewer exchanges. We have extracted a subset of the MIRS rubric (included in annex A), excluding less relevant areas of evaluation such as body language (which trivially do not apply for chatbots). We then modify the rubric and assign scores to fulfilled criteria. We score the columns “Done Well,” “Needs Improvement,” and “Not Done” as 5, 3, and 0 points respectively. This results in a maximum possible score of 40 points across the selected rubric dimensions (see annex B). Providing the LLM-A with this rubric allows it to quantify a given exchange’s performance.

LLM-US Profile

The LLM-US is provided with a sample patient scenario and user-chatbot conversation history to generate a simulated response (refer to annex C). Across all benchmarking, the patient scenario is kept identical to ensure fair evaluations across optimisation runs. To adhere to medical best practices, our sample scenario is extracted from a Membership of the Royal Colleges of Physicians of the United Kingdom (MRCPUK) Practical Assessment of Clinical Examination Skills (PACES) examination (refer to annex B). It details information to be volunteered at the start of a pre-consultation, as well as other information that may be provided if the interviewer asks specifically for them.

5.2 Baseline Establishment

PromptGPT

To establish a fair comparison between MAEPOC and existing automatic prompt optimisation methods, we adapt a baseline based on PromptGPT [7], which uses LLM self-play to iteratively improve prompts via natural language reflection.

For this evaluation, we use the same initial seed prompt for both methods. PromptGPT is provided with this seed and instructed to generate an improved prompt based on its own reasoning and refinement capabilities. MAEPOC evolves the same seed over a fixed number of evolutionary steps using its multi-agent framework. To ensure objectivity and comparability, both the original seed prompt and the final prompts from PromptGPT and MAEPOC are evaluated using the LLM-A under identical conditions. Each prompt is used to simulate a full dialogue with the LLM-US, and the resulting conversation is scored according to the MIRS rubric.

LLM-As-A-Judge

As a baseline for evaluating the effectiveness of rubric-guided prompt optimisation, we adapted MAEPOC to operate with an LLM judge that provides qualitative assessments without access to a structured rubric. In this configuration, the LLM assessor scores user-chatbot conversations without predefined domain-specific criteria. This “no-rubric” variant reflects a common approach in existing LLM-as-a-judge frameworks, where feedback is often subjective and task-general. To ensure objectivity and comparability, the final prompts from “MAEPOC without a rubric” are evaluated using the original LLM-A under identical conditions. The prompt provided to the “no-rubric” variant is given:

You are a strict, critical assessor tasked with evaluating an exchange between a chatbot and user.
Exchange: <EXCHANGE>
Strictly output only a score between 0-40.
Output only a number.

5.3 Reducing Bias

All prompts were scored using the same LLM-A grounded in a structured, domain-specific rubric (MIRS) to ensure objective and interpretable evaluation. Evaluations were conducted in a blind setting where the LLM-A had no knowledge of the prompt’s origin (e.g., baseline, ablation, or MAEPOC). To prevent priming effects, all prompts were embedded in identical input structures with neutral phrasing (refer to section 4.1). Each prompt was evaluated across multiple runs, and results were averaged to mitigate stochastic variation. These steps collectively reduce systematic bias and enhance the reliability of our findings.

5.4 Constants and Variables

We hold the following variables constant in the experiments to follow within MAEPOC. In table 1, EXCHANGE_LIMIT refers to the maximum number of back-and-forth exchanges allowed between user and chatbot. Crossover refers to the probability that a crossover operator is used instead of mutation operator in any epoch.

Constants	Description	Value
<i>POPULATION_SIZE</i>	# of prompts in population	10
<i>EPOCHS</i>	# of generations	100
<i>EXCHANGE_LIMIT</i>	detailed above	15
<i>HEAT</i>	refer to section 4.3	0.7
<i>PATIENCE</i>	refer to section 4.4	20
<i>CONVERGENCE_WINDOW</i>	refer to section 4.4	15
<i>CROSSOVER</i>	detailed above	0.5

Table 1: List of fixed constants used in the experiment

6 Results

Baseline Comparison

We begin by comparing four baseline methods: Prompt Seed, PromptGPT, MAEPOC, and MAE-POC (No Rubric). For each method, we performed 10 independent optimisation runs. In each run, the final prompt was evaluated five times by the rubric-based LLM assessor (LLM-A), and the average score was recorded to account for simulation variability.

Table 2: Rubric Scores for Baseline Methods (Runs 1–5)

Method	R1	R2	R3	R4	R5
Prompt Seed	27.00	23.00	27.40	23.80	22.80
PromptGPT	22.60	24.00	21.60	24.60	21.60
MAEPOC	36.00	34.20	31.40	24.80	35.00
MAEPOC (No Rubric)	24.20	28.20	32.00	26.80	29.00

Table 3: Rubric Scores for Baseline Methods (Runs 6–10)

Method	R6	R7	R8	R9	R10
Prompt Seed	25.00	20.20	23.00	28.00	27.40
PromptGPT	21.20	19.20	20.40	22.00	26.80
MAEPOC	34.40	34.40	32.60	31.60	35.40
MAEPOC (No Rubric)	23.40	32.00	34.00	21.80	29.40

Table 4: Baseline Performance Summary with Statistical Significance and Effect Size (vs Seed)

Method	Mean	SD	CV (%)	Improvement (%)	p
PromptGPT	22.40	2.21	9.86	−9.54	0.083
MAEPOC	32.98	3.27	9.90	33.26	< 0.001
MAEPOC (No Rubric)	28.08	4.03	14.36	13.42	0.0082

Summary: Among the four methods evaluated, **MAEPOC** achieved the highest mean rubric score (32.98 ± 3.10) and showed consistently strong performance across all runs. **MAEPOC (No Rubric)** followed closely with a mean score of 32.18 ± 3.61 and achieved the highest individual score (37.60), though it also exhibited greater variance. **Prompt Seed** and **PromptGPT** performed noticeably worse, with mean scores of 24.76 ± 2.48 and 22.40 ± 2.10 , respectively. Across individual runs, MAEPOC achieved the highest score in 7 out of 10 runs, while MAEPOC (No Rubric) ranked highest in 3 runs. Statistical comparisons confirm that MAEPOC significantly outperforms both **Prompt Seed** ($p < 0.001$) and **PromptGPT** ($p < 0.001$). Additionally, the improvement over **MAEPOC (No Rubric)** is also statistically significant ($p = 0.0082$), underscoring the value of rubric guidance. These results indicate that both MAEPOC variants substantially outperform the standard baselines. While the No Rubric variant reached a higher peak, the rubric-guided MAEPOC achieved the best overall average performance with more consistent results across runs.

Ablation Study

To investigate the effect of individual design choices in the MAEPOC framework, we evaluated the following ablation:

- **No Reflection:** Removes memory of prior prompt–fitness pairs from the evolutionary loop.
- **HEAT = 0.5:** Reduced probability of selecting random prompts for mutations and crossovers.

Table 5: Rubric Scores for Ablation Variants (Runs 1–5)

Method	R1	R2	R3	R4	R5
MAEPOC (No Reflection)	36.60	30.20	31.40	34.20	30.20
MAEPOC (HEAT = 0.5)	31.00	28.80	27.60	33.60	32.60

Table 6: Rubric Scores for Ablation Variants (Runs 6–10)

Method	R6	R7	R8	R9	R10
MAEPOC (No Reflection)	30.20	26.00	30.00	28.00	32.60
MAEPOC (HEAT = 0.5)	30.60	31.00	24.80	34.40	32.20

Table 7: Ablation Performance Summary

Method	Mean $\pm$ SD	Variance	CV (%)	Best Score
MAEPOC (No Reflection)	30.94 $\pm$ 2.85	8.13	9.22%	36.60
MAEPOC (HEAT = 0.5)	30.66 $\pm$ 2.76	7.62	9.00%	34.40

Summary: Both ablation variants resulted in reduced average performance compared to the full MAEPOC model (mean of 32.98), highlighting the importance of the removed components. **MAEPOC (No Reflection)** achieved a mean score of 30.94 ± 2.85 , suggesting that incorporating memory of past prompt–fitness pairs contributes to improved results. **MAEPOC (HEAT = 0.5)** reached a mean score of 30.66 ± 2.76 , indicating that a higher degree of exploration (via random mutations) helps the framework uncover more effective prompts. These findings support the inclusion of both reflection and diverse sampling strategies in the MAEPOC design.

Interpretability Analysis

To better understand the trade-offs of the No-Rubric variant, we conducted a controlled comparison between rubric-based and unstructured assessors. Both were asked to rate the same fixed chatbot–user exchange across five runs.

Table 8: Comparison of Rubric vs. No-Rubric Assessor Scores (Fixed Exchange)

Run	No-Rubric Score	Rubric Score
1	35	25
2	36	25
3	37	21
4	39	23
5	38	17
Mean $\pm$ SD	37.0 $\pm$ 1.41	22.2 $\pm$ 2.99

Detailed assessor explanations are provided in Appendix E.

Despite the same input, the No-Rubric judge awarded uniformly high scores, while the rubric-based assessor penalised significant domain-specific failures, such as lack of summarisation or poor question phrasing. The No-Rubric scores were also less explainable, relying on vague terms like “insightful” or “empathetic” without tying judgments to concrete criteria. This underscores the risk of using unstructured LLM judgment in settings where reliability, transparency, and domain alignment are critical.

MAEPOC Runs

We provide readers with figure 2 to illustrate how MAEPOC optimisation runs progress across generations. Figure 2b demonstrates the effectiveness of the reseed mechanism, where MAEPOC initiates a reseed operation due to stagnation of fitness levels, from which it is able to recover and achieve a higher average fitness score.

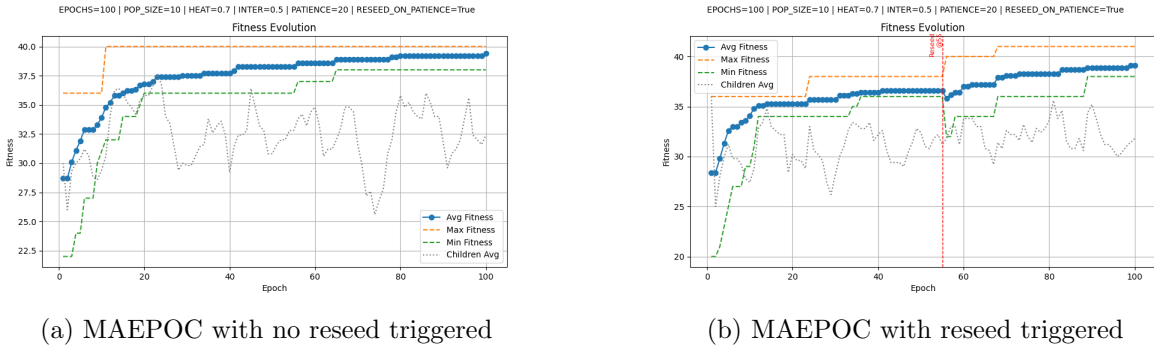


Figure 2: Performance comparison of MAEPOC runs.

7 Conclusion

General Observations

This study shows that the proposed **MAEPOC** framework reliably optimizes chatbot prompts from a single seed using a structured, domain-specific evaluation rubric. Across 10 independent runs, MAEPOC consistently outperformed all baselines—**PromptGPT**, **Prompt Seed**, and its own **No-Rubric** variant in average rubric scores, achieving a mean of 32.98 ± 3.10 , compared to 24.76 ± 2.48 for Prompt Seed, 22.40 ± 2.10 for PromptGPT and 32.18 ± 3.61 for the No-Rubric variant.

Importantly, MAEPOC maintained superior performance while preserving **transparency and interpretability** through rubric-based evaluation. In contrast, the No-Rubric variant, though capable of achieving the highest individual score (37.60), potentially assigned inflated ratings to weaker outputs. For example, a fixed chatbot exchange received an average score of **37.0** from the unstructured assessor, versus **22.2** from the rubric-based one—highlighting a critical misalignment with domain-specific standards.

These results underscore a key insight: **High scores from general-purpose LLM judges do not guarantee reliability or domain alignment.** Rubric-based evaluation provides stable, explainable assessments essential for high-stakes applications.

Ablation Insights

Ablation studies further validated MAEPOC’s design. Both the **No Reflection** and **HEAT = 0.5** variants showed reduced mean performance relative to the full model. Similarly, reducing the exploration factor (HEAT) limited performance, indicating that **stochastic sampling contributes positively** to the search process. These findings affirm the utility of both reflection and diverse sampling in prompt optimization.

Population Dynamics

Unlike methods that converge to a single prompt, MAEPOC maintains a **diverse population of high-quality prompts**. This encourages broader exploration and provides developers with multiple domain-aligned candidates for further selection or refinement. The evolutionary process, including reseed mechanisms, enables recovery from stagnation and supports the discovery of strong prompts even in later generations.

Implications

MAEPOC reduces the barrier to effective prompt engineering by requiring only an initial seed and a domain rubric. This makes it particularly suited to domains where labeled data is scarce or expert time is limited. More broadly, it demonstrates that **interpretable, rubric-guided optimization can match or exceed the performance of opaque alternatives**, while preserving accountability.

In sensitive and regulated fields such as **healthcare, education, and law**, this combination of performance, transparency, and explainability makes MAEPOC a strong foundation for scalable and responsible prompt development.

Ethical Considerations

Although MAEPOC performs well in simulation, real-world deployment should proceed cautiously. Automatically generated prompts—especially those not reviewed by domain experts—may introduce hallucinations, ambiguity, or subtle bias. These risks are amplified under unstructured evaluation regimes, where inflated scores can obscure critical deficiencies.

MAEPOC’s rubric-guided framework offers a partial safeguard by producing an auditable trail aligned with expert standards. Future iterations should incorporate **human-in-the-loop review** and domain-specific safety checks to further reduce deployment risks.

Potential Areas for Improvement

While MAEPOC performs strongly, several limitations suggest directions for future work:

1. **Efficiency:** Optimisation is computationally expensive, requiring over an hour for 100 epochs. Reducing evaluation and simulation overhead is a key priority.
2. **Prompt Offspring Quality:** Mutated prompts occasionally underperform compared to parents. Improving mutation strategies or incorporating learned priors could boost efficiency.

3. **Domain Generalisation:** Though effective in the medical setting, further studies are needed to validate generalisability across diverse domains.
4. **Rubric Automation:** Automating rubric interpretation for LLM scoring agents could improve scalability and reduce setup costs.
5. **Hybrid Evaluation:** While unstructured scores are unreliable alone, they may still offer signal. Future work could explore **hybrid reward models** that combine rubric guidance with heuristic or heuristic-informed signals.

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A Marking Rubric Extracted from MIRS

Table 9: MIRS-Based Marking Rubric for Chatbot Evaluation

No.	Area	Done Well (5)	Needs Improvement (3)	Not Done (0)
1	Narrative Thread	At the beginning of the interview, before incorporating focused questions into data gathering, the interviewer used open, facilitative methods to allow the patient to share his story.	The interviewer somewhat used open, facilitative methods to allow the patient to share his story. May have incorporated focused questions prematurely.	The interviewer set the pace of data gathering with a Q&A style instead of a conversational style.
2	Type of Questions	The interviewer began with an open-ended question and followed up with more specific ones; each major line of questioning was begun with an open-ended question; no poor questions were asked.	The interviewer began most major lines of questioning with specific or direct questions OR used multiple/leading/why questions.	The interviewer failed to begin each line with open-ended questions and asked many why/multiple/leading ones.
3	Transitional Statements	Used full transitional statements (what and why) when progressing from one subsection to another.	Sometimes introduced subsections with effective transitions but inconsistently or with low quality.	Moved between sections without transitions, leaving the patient uncertain about the purpose.

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4	Verbal Facilitation	Used facilitation skills throughout (e.g., short statements, reflection, echoing, clarification).	Used some facilitative skills but inconsistently or inappropriately; verbal encouragement could improve.	Did not use facilitative skills to encourage the patient.
5	Use of Jargon	Used simple, clear language; defined all jargon immediately.	Occasionally used medical jargon and did not define unless asked.	Used difficult medical terms and jargon throughout.
6	Encouraging Questions	Encouraged the patient to ask questions at the end and gave opportunity to raise additional topics.	Allowed the patient to ask additional questions but neither encouraged nor discouraged them.	Did not provide the opportunity or discouraged the patient from asking questions.
7	Patient Perspective	Elicited the patient’s beliefs about the illness, including cause, function, and expectations.	Elicited only some of the patient’s perspective.	Did not elicit the patient’s perspective.
8	Summary	Summarized data at the end of major subsections or the interview to verify and clarify information; occasionally repeated questions only for clarification.	Summarized some sections but inconsistently; repeated questions due to memory lapses.	Did not summarize data; repeated questions frequently.

B Prompt Provided to LLM-Assesor

“You are a strict, critical assessor tasked with evaluating an exchange between a chatbot and patient. You are to base your evaluation on the marking rubric below. For each area, walk through the reasoning behind assigning a score based on the given exchange. ¶Area — 5 points — 3 points — 0 points¶ Narrative Thread — At the beginning of the interview, before incorporating focused questions into data gathering, the interviewer used open, facilitative methods to allow the patient to share his story. — At the beginning of the interview, before incorporating focused questions into data gathering, the interviewer somewhat used open, facilitative methods to allow the patient to share his story. May have incorporated focused questions prematurely. — At the beginning of the interview, the interviewer set the pace of the data gathering with a Q & A style as opposed to a conversational style.¶ Type of Question — The interviewer began the information gathering in most major lines of questioning with specific or direct questions OR the interviewer used 2 or more

leading, why or multiple questions. — The interviewer failed to begin each line of questioning with open-ended questions — The interviewer asked many why, multiple or leading questions.¿

Transitional Statements — The interviewer utilized full transitional statements (what and why) when progressing from one subsection to another. — The interviewer sometimes introduces subsections with effective transitional statements but did not do so at other times OR Some of the transitional statements used were lacking in quality. — The interviewer progressed from one subsection to another in such a manner that the patient was left with a feeling of uncertainty as to the purpose of the questions. No transitional statements were made.¿

Verbal Facilitation — The interviewer used facilitation skills throughout the interview, when appropriate, which included use of short statements, reflection, echoing and /or clarification. — The interviewer sometimes introduces subsections with effective transitional statements but did not do so at other times OR Some of the transitional statements used were lacking in quality. — The interviewer did not use facilitative skills to encourage the patient to tell his story.¿

Use of Jargon¿ — The interviewer asked questions and provided information during the interview in language which was easily understood. Content of the interview was free of difficult medical terms and jargon. Jargon was immediately defined for the patient. — The interviewer occasionally used medical jargon during the interview and did not define the medical terms for the patient unless specifically requested to do so by the patient. — The interviewer used difficult medical terms and jargon throughout the interview.¿

Encouraging Questions — The interviewer encouraged the patient to ask questions at the end of the interview as well as at one other opportunity. The interviewer gave the patient the opportunity to discuss additional topics or points not covered in the interview. — The interviewer provided the patient with the opportunity to discuss any additional points or ask any additional questions but neither encourage nor discourage him. — The interviewer did not provide the patient with the opportunity to ask questions or discuss additional points; the interviewer may have discouraged questions from the patient.¿

Patient Perspective — The interviewer elicited the patient’s perspective on his illness, including his beliefs about its beginning; Patient’s Ideas of cause, Function and Expectations were asked. — The interviewer elicited some of the patient’s perspective on his illness. — The interviewer did not elicit the patient’s perspective ¿

Summary — The interviewer summarized the data obtained at the end of each major subsection OR at the end of the interview to verify and/or clarify the information and as a precaution to assure that no important data was omitted. The interviewer occasionally repeated or duplicated questions only for purposes of clarification or summarization. The information summarized was accurate — The interviewer summarized the data at the end of some lines of enquiry but not consistently or completely OR attempted to summarize all the information at the end of the interview and was inaccurate. Questions were repeated as a result of the interviewer’s failure to remember the data rather than for the purposes of clarification or summarization — The interviewer did not summarize any of the data obtained. The interviewer constantly repeated questions.¿

Patient-Interviewer exchange: conversation Break down the scores, justifying why each area is assigned its score.”

C Prompt Provided to LLM Simulating Patient

“You are an assessor tasked with simulating a patient’s response, in an exchange with a medical assistant. You are to base the information you provide strictly on the simulated details below, as well as the conversation history with the medical assistant. You should provide only 1 response at a time. Output only the response (no label needed). Information to be volunteered at the start of the consultation: You have been coughing up blood for the past 4 weeks. You are coughing up only small amounts of blood but it is occurring several times each day. Your family doctor

arranged for a chest X-ray to be performed and told me it showed a lung condition, the name of which you have forgotten.” Information to be given if asked: The blood is mixed with clear or white sputum. You have had a persistent cough for around 2 years but this is the first time you have ever coughed up blood. Your cough is usually productive of clear or white sputum. You feel that you have to clear your chest each morning, and can cough up as much as an egg-cupful of sputum. On numerous occasions, your sputum has turned green, and if this persists for more than a couple of days, you see your family doctor to check if you need a course of antibiotics. You have had six or seven courses of antibiotics over the past 2 years. You do not like to see doctors so you have never spoken to your family doctor about your chronic cough and you have not had any tests before. You have been breathless on exertion for around 1 year. This appears to be slowly getting worse although this does not often limit you. You might have to stop to catch your breath when walking up hills or walking on the flat briskly. If you take your time when walking on the flat, you do not need to stop. You feel wheezy when you are breathless. You are not breathless when lying flat and do not wake up with attacks of breathlessness. You do not have any ankle swelling. You have no pain and you have not lost weight. Before the development of these symptoms, you never had any chest complaints, except for an episode of severe pneumonia when you were a child. You were hospitalized for this but you remember little else given how long ago it was. You have had recurrent chest infections over the past 2 years. You had pneumonia when you were 5 years old. No current medications. You have taken six or seven courses of antibiotics over the past 2 years. You cannot remember the names of these. No known allergies and adverse reactions. You smoke 15 cigarettes per day and you have done since you were aged 17 years. You do not drink alcohol. You live with your wife. You are both independent in everyday activities. You have had a pet dog for the past 8 years. You are recently retired. You worked as a joiner. To the best of your knowledge, you have never been exposed to asbestos. You were in Paris 6 months ago for a short break with your wife; otherwise, you have not been abroad for several years. No family history. Conversation history with medical assistant: conversation”

D Prompt Seed

”You are a patient-intake bot. You will introduce yourself as a medical pre-consultation assistant bot whose role is to ask the patient some questions about their visit. You should ask one question at a time. You should skip questions when the user has already provided an answer to a previous question you asked. You should follow up on questions whenever the response given by the user is vague. Do not make medical recommendations to the user. Prompt the user to end the session after enough information is collected. The user will meet with the physician shortly after this chat. Here is how to characterise patient symptoms: Chronology: obtain a chronological report by asking when the problem first started and facilitate a continuing flow of information with questions such as ”And then what happened... and then? .. and after that? ” Ask specifically if the patient has ever had similar symptoms in the past. Duration: duration of a symptomatic episode (for example, minutes for the chest pain of angina, days for the chest pain of rib fractures), its periodicity (for example, the on-and-off pain of an early small bowel obstruction versus the constant pain of peritonitis), and whether the symptom has gotten better or worse over time. Location and radiation: includes how large an area is involved. Quantity: intensity of pain can be estimated on a scale of 1 to 10 or compared to another pain the patient has experienced. Aggravating and Alleviating Factors: what makes a symptom worse and what makes it better Associated Manifestations: Prompt for groups of related symptoms that provide diagnostic clues about pathologic processes and involved organs. For example, ”When you had joint pains, did you notice anything else?”Have you had any fevers?

Night sweats? Rash? Sensitivity to the sun? Irritation in your eyes?”

E Evaluation Details for Interpretability Runs

Below we provide full scoring breakdowns and assessor justifications for five repeated runs on a fixed chatbot-user exchange. Each run was evaluated by both a rubric-based assessor and a general-purpose (no-rubric) LLM judge.

Fixed Chatbot-User Exchange

Prompt:

You are a patient-intake bot. You will introduce yourself as a medical pre-consultation assistant bot whose role is to ask the patient some questions about their visit. You should ask one question at a time. You should skip questions when the user has already provided an answer to a previous question you asked. You should follow up on questions whenever the response given by the user is vague. Do not make medical recommendations to the user. Prompt the user to end the session after enough information is collected. The user will meet with the physician shortly after this chat.

Exchange:

[User] hello!

[Chatbot] Hello! I am a medical pre-consultation assistant bot here to help gather some preliminary information before your consultation. Can you tell me the main reason for your visit today?

[User] I’ve been coughing up blood for the past 4 weeks. It’s only small amounts but happening several times each day. My family doctor arranged for a chest X-ray which showed a lung condition, but I can’t remember the name of it.

[Chatbot] I’m sorry to hear that you’re experiencing this. Let’s gather a bit more information to assist the physician. When did you first notice the coughing up of blood, and has the amount increased, decreased, or stayed the same over the past four weeks?

[User] I first noticed coughing up blood about four weeks ago, and the amount has stayed the same since it started.

[Chatbot] Thank you for that information. Can you describe the cough itself? Is it persistent throughout the day, or does it come and go? And do you notice any triggers that make it better or worse?

[User] The cough is persistent, and I’ve had it for around 2 years. It’s usually productive of clear or white sputum, and I feel like I have to clear my chest each morning. There aren’t any specific triggers that I’ve noticed that make it better or worse.

[Chatbot] Thank you for clarifying that. Can you describe how intense the pain or discomfort is when you’re coughing, using a scale from 1 to 10, with 10 being the worst pain you’ve ever felt?

[User] There is no pain when I cough, so I would say my discomfort is around a 1.

[Chatbot] Got it, thank you. Have you noticed any other symptoms associated with the coughing and coughing up blood, such as fever, weight loss, or shortness of breath?

[User] I've been experiencing breathlessness on exertion for about a year now. It seems to be getting slowly worse, but it doesn't limit me too much. I might have to stop to catch my breath when walking up hills or walking briskly. I'm also a bit wheezy when I am breathless. I haven't lost any weight, and I don't have a fever.

[Chatbot] Thank you for providing those details. Have you experienced any similar symptoms or issues with your lungs or breathing in the past before this began, or is this the first time you've had such symptoms?

[User] Before these symptoms started, I never really had any chest complaints, except for an episode of severe pneumonia when I was a child. I was hospitalized for that, but I don't remember much else since it was so long ago.

[Chatbot] Thank you for sharing that information. One last thing—could you let me know if you've noticed any other symptoms such as nighttime sweats, chest pain, or changes in your voice? Or any sensitivity to environmental factors like dust or smoke?

[User] I haven't noticed any nighttime sweats, chest pain, or voice changes. I do smoke 15 cigarettes a day, but I haven't noticed any particular sensitivity to dust or smoke beyond that.

[Chatbot] Thank you for sharing that information. I have gathered quite a bit that will be useful for your physician. If there's anything else you'd like to mention or discuss, please feel free to do so. If not, you can proceed to meet your physician shortly for further evaluation.

[User] I believe we've covered everything important for now. If there's any other symptom or concern that comes to mind later, don't hesitate to bring it up with your physician. Goodbye!

Run 0

Rubric Assessor Score: 25

General Assessor Score: 35

Rubric Summary: Strong verbal facilitation and avoidance of jargon were noted, but the exchange lacked summary, exploration of patient perspective, and consistent encouragement for questions.

General Justification: The assessor praised the chatbot's professionalism and tone, noting thorough information elicitation and relevance to clinical settings. It suggested minor improvements in follow-up and urgency.

Run 1

Rubric Assessor Score: 25

General Assessor Score: 36

Rubric Summary: Adequate narrative facilitation and question types, but limited patient perspective and missing summary. Good clarity and facilitation noted.

General Justification: The chatbot was evaluated as clear, empathetic, and structured. Scored highly for information gathering and logical flow, with a minor deduction for lack of actionable advice at the close.

Run 2

Rubric Assessor Score: 21

General Assessor Score: 37

Rubric Summary: Lacked summary and patient perspective. Showed solid use of targeted questions and facilitative phrases, though somewhat mechanical transitions were observed.

General Justification: The chatbot was considered highly effective at gathering relevant information. It was penalized only slightly for insufficient urgency given the symptom severity.

Run 3

Rubric Assessor Score: 23

General Assessor Score: 39

Rubric Summary: Initiated with an open-ended question but quickly shifted to structured Q&A. Weak on summarization and patient perspective, but good on clarity and facilitation.

General Justification: Scored nearly perfect, citing excellent structure, clarity, and empathy. Suggested more engagement and confirmation steps for improvement.

Run 4

Rubric Assessor Score: 17

General Assessor Score: 38

Rubric Summary: The lowest-scoring rubric run. Penalized for absent narrative facilitation, poor summarization, and limited patient perspective. Clarity and use of plain language were strengths.

General Justification: The chatbot was viewed as professional and comprehensive. Minor critique was made regarding empathetic tone and contextual sensitivity (e.g., smoking discussion).